Optimization for distributed training

ONE LOVE. ONE FUTURE.

1

Recap: large models

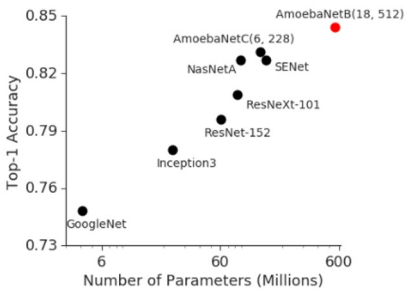
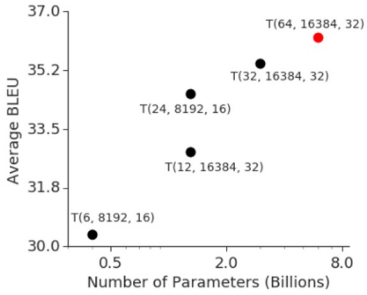


Image Classification
ImageNet



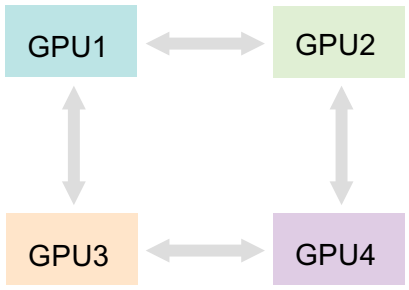
Machine Translation
average over WMT

Source: https://arxiv.org/abs/1811.06965

2

Recap: Ring allreduce

Bonus quest: you can only send data between **adjacent** gpus



Ring topology

Image: graphcore ipu server

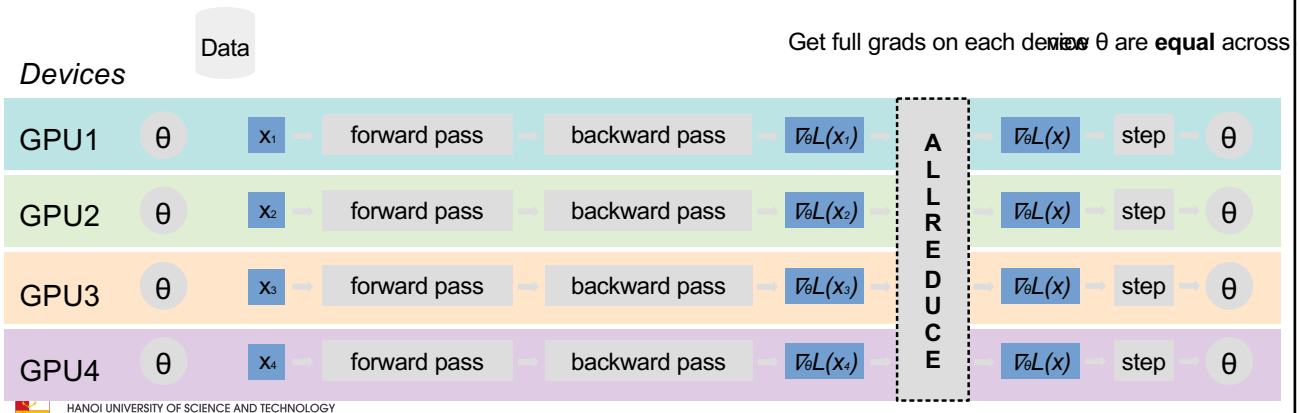
Answer & more: tinyurl.com/ring-allreduce-blog

3

Recap: All-Reduce SGD

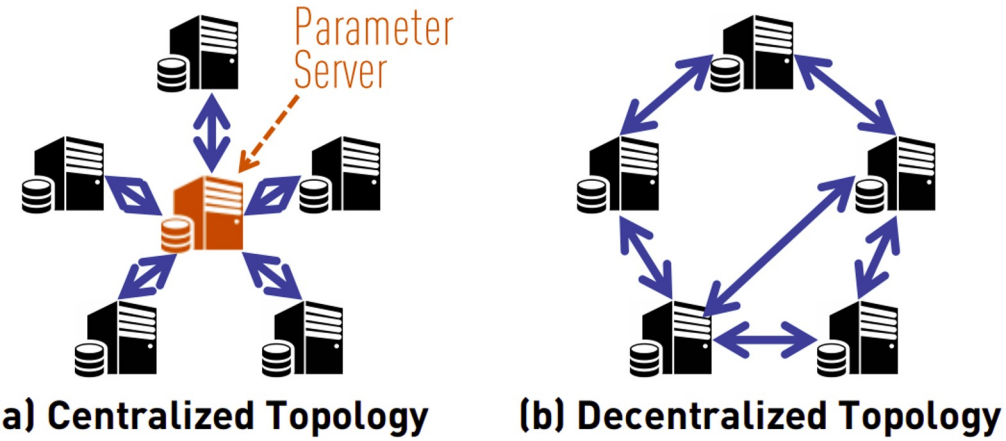
arxiv.org/abs/1706.02677

Idea: get rid of the host, each gpu runs its own computation
Q: why will weights be equal after such step?



4

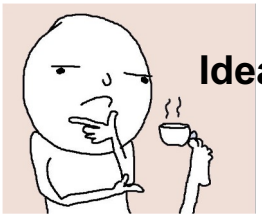
Recap: Decentralized SGD



- **Q:** What if a model is larger than GPU?
-

- **Q:** What if a model is larger than GPU?
easy mode: cannot fit the right batch size
  - hard mode: cannot fit a single sample
    - expert mode: not even parameters!

- **Q:** What if a model is larger than GPU?
easy mode: cannot fit the right batch size
  - hard mode: cannot fit a single sample
    - expert mode: not even parameters!



- **Q:** What if a model is larger than GPU?
  easy mode: cannot fit the right batch size
  - hard mode: cannot fit a single sample
    - expert mode: not even parameters!

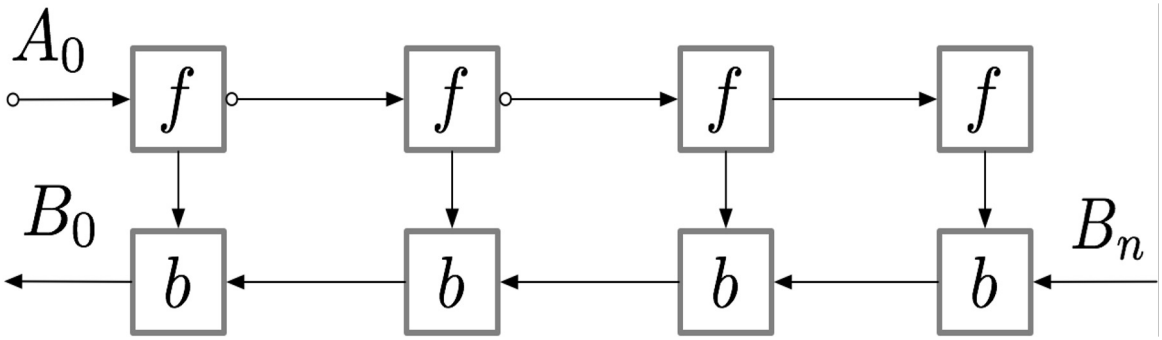
Solution: accumulate grads from several training batches

```
[ ]  1 optimizer.zero_grad()
     2 for i in range(B):
     3   loss = model(**next_batch())
     4   (loss / B).backward()
     5 optimizer.step()
```


- **Q:** What if a model is larger than GPU?
  easy mode: cannot fit the right batch size
- **hard mode:** cannot fit one training sample
  - expert mode: not even parameters!

Gradient checkpointing

aka rematerialization

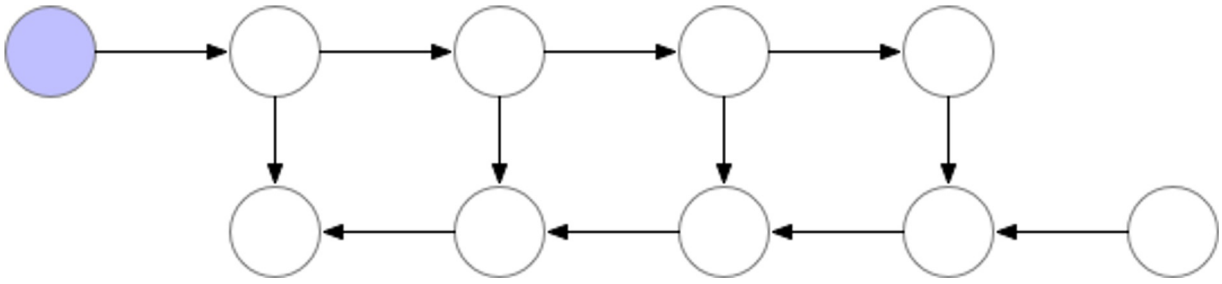


Paper (DL): arxiv.org/pdf/1604.06174.pdf
TF: github.com/cybertronai/gradient-checkpointing
Pytorch: pytorch.org/docs/stable/checkpoint.html

11

Gradient checkpointing

Normal backprop

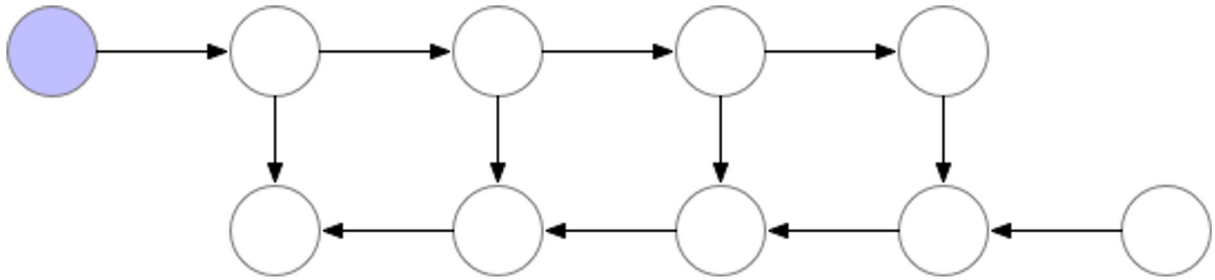


Paper (DL): arxiv.org/pdf/1604.06174.pdf
TF: github.com/cybertronai/gradient-checkpointing
Pytorch: pytorch.org/docs/stable/checkpoint.html

12

Gradient checkpointing

Full rematerialization



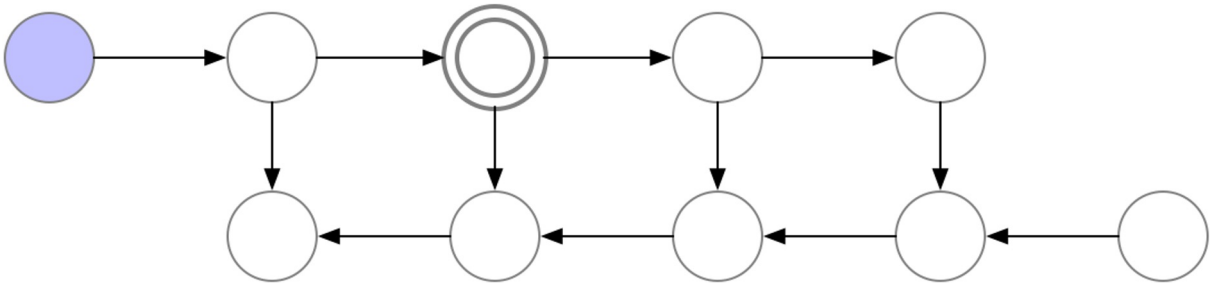
Paper (DL): arxiv.org/pdf/1604.06174.pdf
TF: github.com/cybertronai/gradient-checkpointing
Pytorch: pytorch.org/docs/stable/checkpoint.html

13

Gradient checkpointing

Single checkpoint

checkpoint

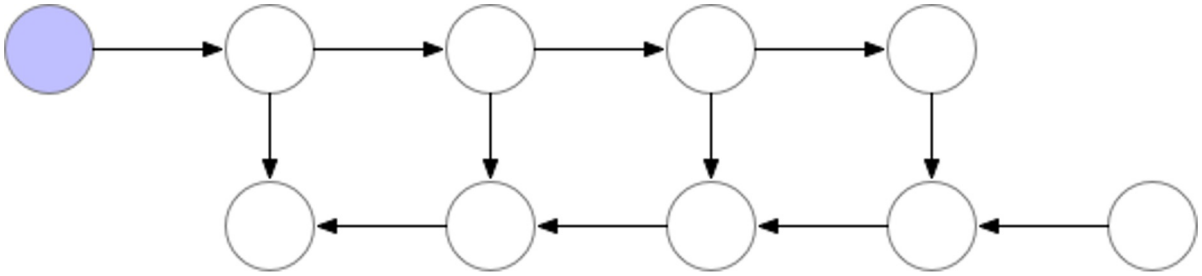


Paper (DL): arxiv.org/pdf/1604.06174.pdf
TF: github.com/cybertronai/gradient-checkpointing
Pytorch: pytorch.org/docs/stable/checkpoint.html

14

Gradient checkpointing

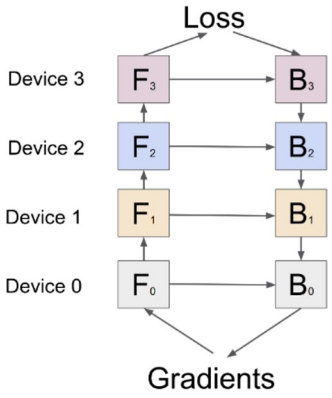
Single checkpoint



Paper (DL): arxiv.org/pdf/1604.06174.pdf
TF: github.com/cybertronai/gradient-checkpointing
Pytorch: pytorch.org/docs/stable/checkpoint.html

- **Q:** What if a model is larger than GPU?
  easy mode: cannot fit batch size 1
- **expert mode:** not even parameters!

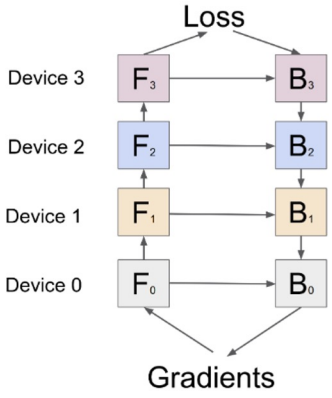
Model-parallel training



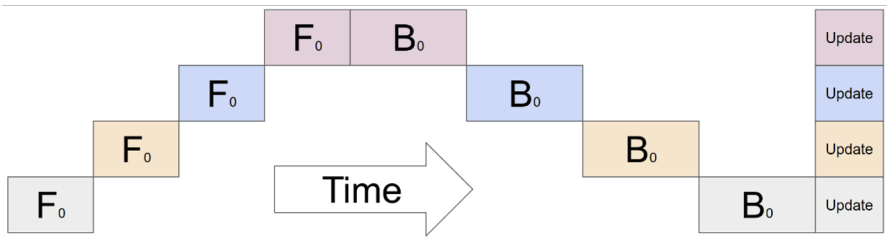
- **Q:** What if a model is larger than GPU?

17

Model-parallel training



- **Q:** What if a model is larger than GPU?



model size: O(N)
throughput: O(1)

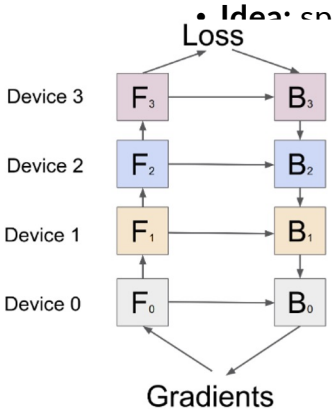
Q: Can we go faster?

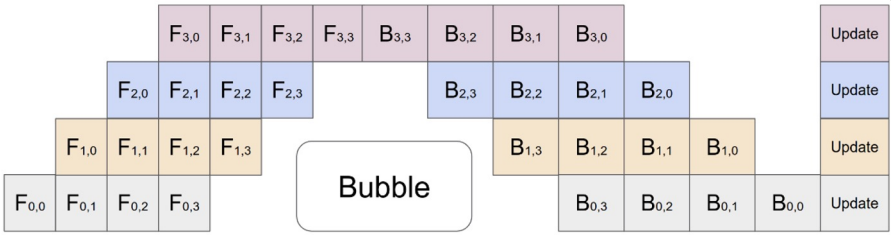
18

Pipelining

GPipe: arxiv.org/abs/1811.06965 – good starting point, *not* the 1st paper

- **Idea:** split data into micro-batches and form a pipeline (right)





model size: O(n)
throughput: O(n) – with caveats

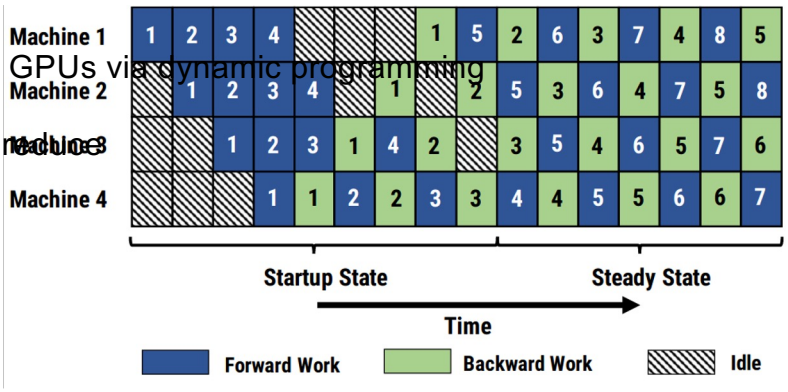
Q: Even faster?

Pipeline-parallel training

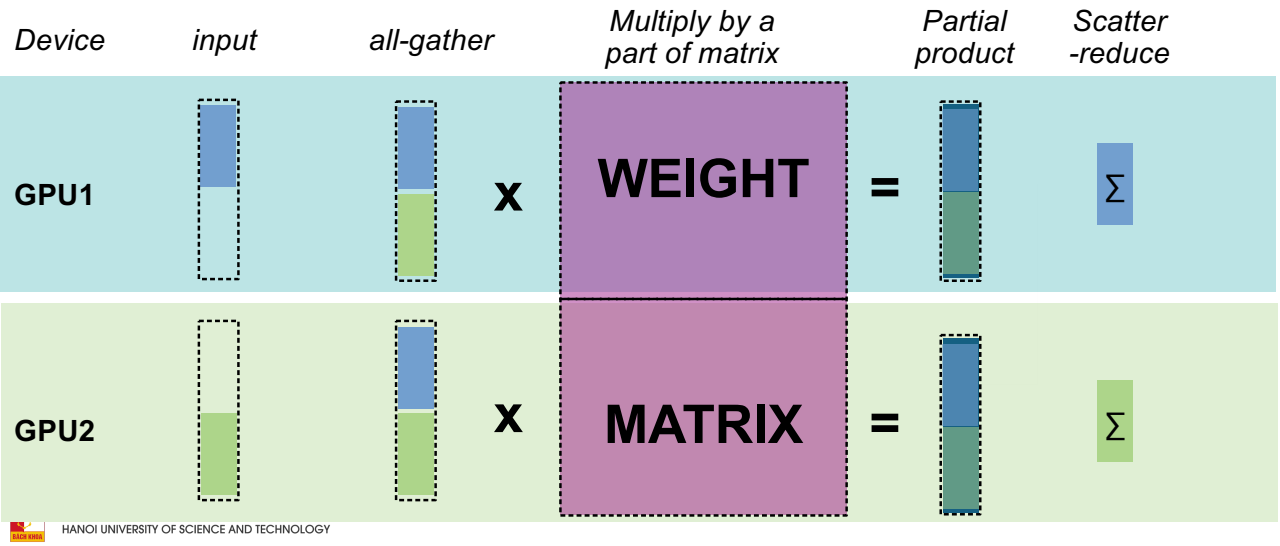
PipeDream: arxiv.org/abs/1806.03377

Idea: apply gradients with every microbatch for maximum throughput

Also neat:
- Automatically partition layers to GPUs via dynamic programming
- *Store k* past weight versions to reduce gradient staleness
- Aims at high latency



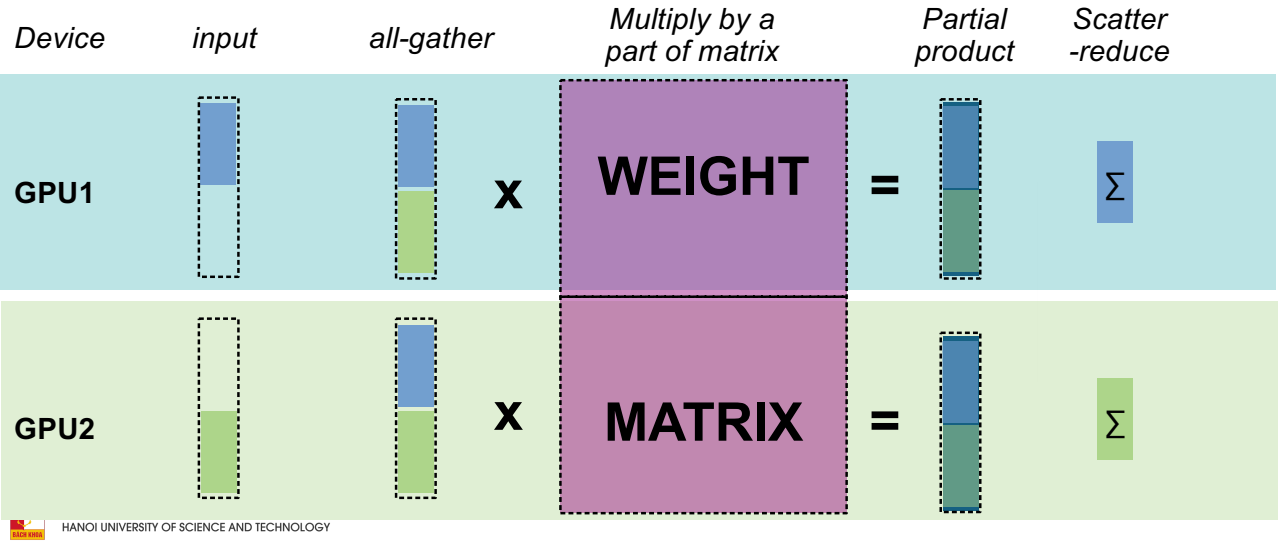
Tensor-parallel training



21

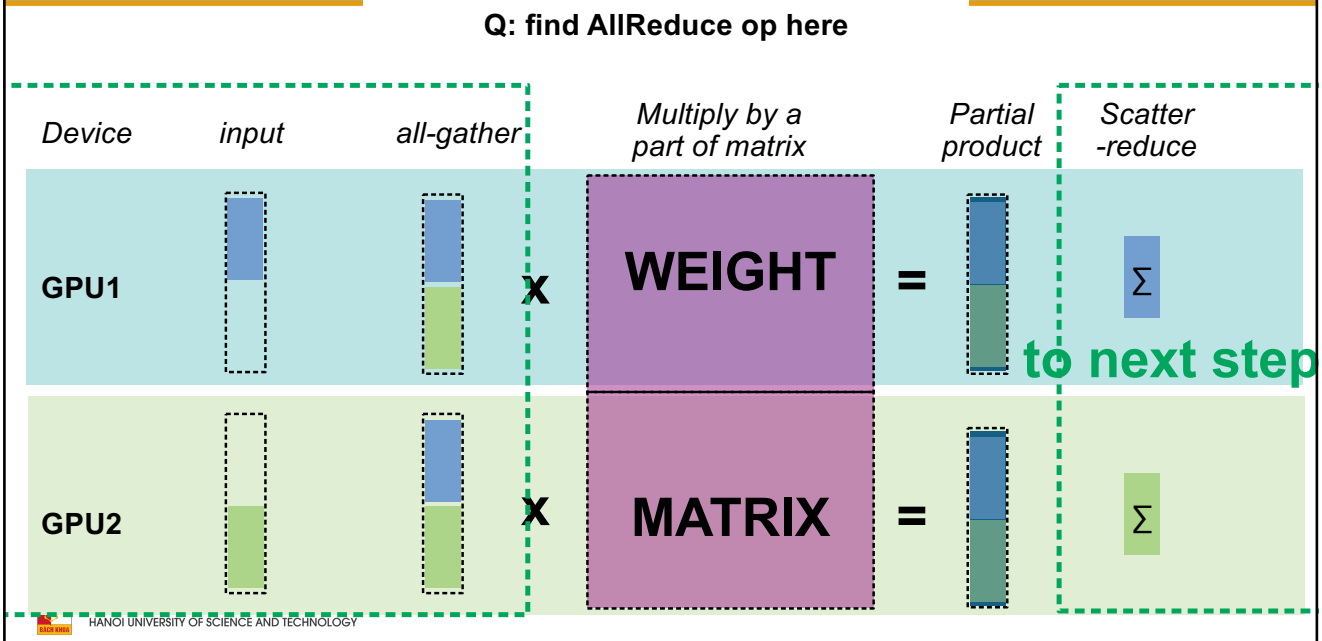
Tensor-parallel training

Q: find AllReduce op here



22

Tensor-parallel training



Model-parallel

+ model larger than GPU
+ faster for small
* typical size: 2-8 gpus
- model partitioning is tricky
  tensor parallelism is easier, but requires ultra low latency
- latency is critical, go buy nvlink
  except for PipeDream
- often combined with gradient checkpointing

Tutorials:

- Simple pipelining in PyTorch – tinyurl.com/pytorch-pipelining
- Distributed model-parallel with torch RPC - https://tinyurl.com/torch-rpc
- Automatic tensor parallelism pip install tensor_parallel

Model parallel

- + model larger than GPU
- + faster for small
- * typical size: 2-8 gpus
- - model partitioning is tricky
- tensor parallelism is easier, but requires ultra low latency
- - latency is critical, go buy nvlink
- *except for PipeDream*
- *- often combined with gradient checkpointing*

- **Tutorials:**

.Simple pipelining in PyTorch – tinyurl.com/pytorch-pipelining
.Distributed model-parallel with torch RPC - https://tinyurl.com/torch-rpc
.Automatic tensor parallelism pip install tensor_parallel

Q: what if you have 1024 GPUs, but the model fits on 8?

Model parallel

- + model larger than GPU
- + faster for small
- * typical size: 2-8 gpus
- - model partitioning is tricky
- tensor parallelism is easier, but requires ultra low latency
- - latency is critical, go buy nvlink
- *except for PipeDream*
- *- often combined with gradient checkpointing*

- **Tutorials:**

.Simple pipelining in PyTorch – tinyurl.com/pytorch-pipelining
.Distributed model-parallel with torch RPC - https://tinyurl.com/torch-rpc
.Automatic tensor parallelism pip install tensor_parallel

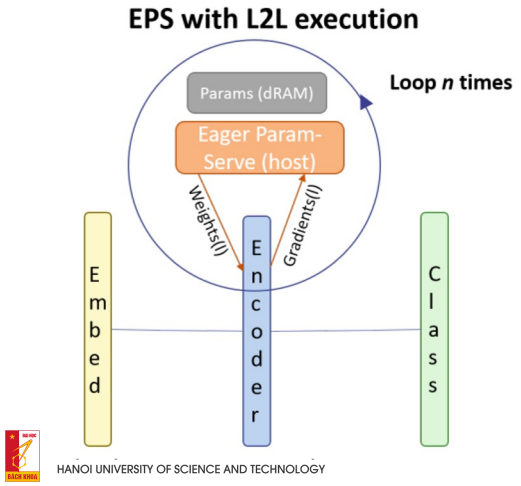
Large-scale training: combine model- and data-parallel

Case study: DeepSpeed

Source: microsoft

Memory offloading

L2L: https://arxiv.org/abs/2002.05645



- Initialize all layers on CPU
- Move k layers at a time to GPU
- Remove layers after computation

- Fetch k+1-st layer while k-th runs
- Still 20-50% overhead

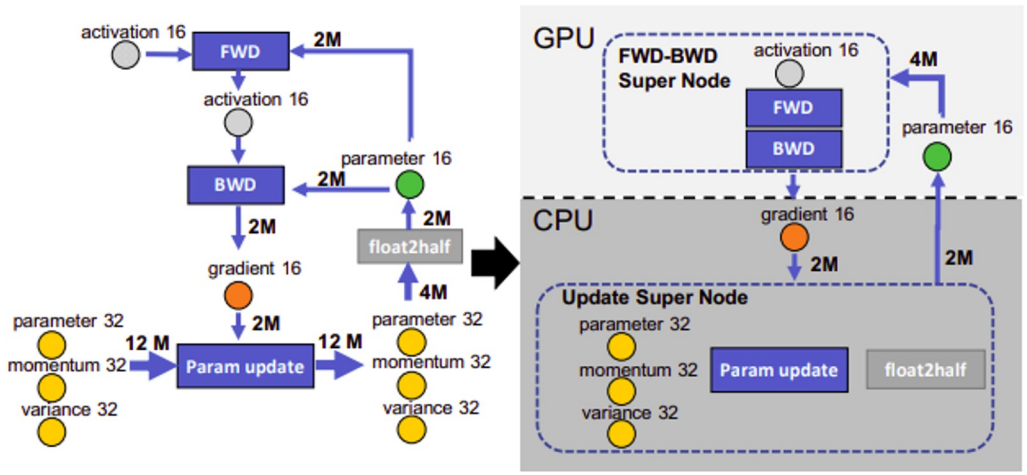
Memory offloading

L2L: https://arxiv.org/abs/2002.05645

METHOD	UBATCH SIZE	DEVICE BATCH SIZE	#LAYER	#PARAMETERS	MEMORY (GB)
BASELINE	2	2	24	300 MILLION	9.23
BASELINE	2	**2**	**48**	600 MILLION	**OOM**
L2L-STASH ON GPU	64	64	24	300 MILLION	5.22
L2L-STASH ON GPU	**64**	**64**	**48**	600 MILLION	**6.76**
L2L-STASH ON GPU	**64**	**64**	**96**	1.2 BILLION	**9.83**
L2L-STASH ON CPU	64	64	24	300 MILLION	3.69
L2L-STASH ON CPU	**64**	**64**	**96**	1.2 BILLION	**3.69**
L2L-STASH ON CPU	**64**	**64**	**384**	4.8 BILLION	**3.69**

Memory offloading

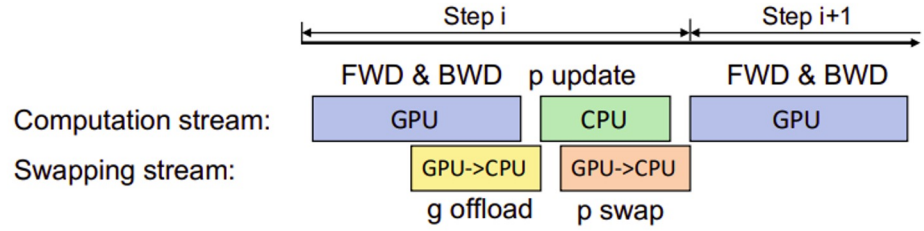
ZeRO-offload: https://arxiv.org/abs/2101.06840



Memory offloading

ZeRO-offload: https://arxiv.org/abs/2101.06840

- Offload **in parallel** with computation
- Use gradient checkpointing
- Delayed parameter update



Memory offloading

ZeRO-offload: https://arxiv.org/abs/2101.06840

- Offload in parallel with computation
- Use gradient checkpointing
- **Delayed parameter update**

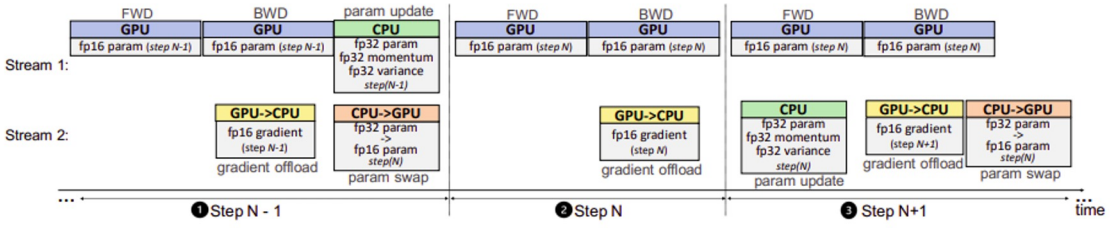
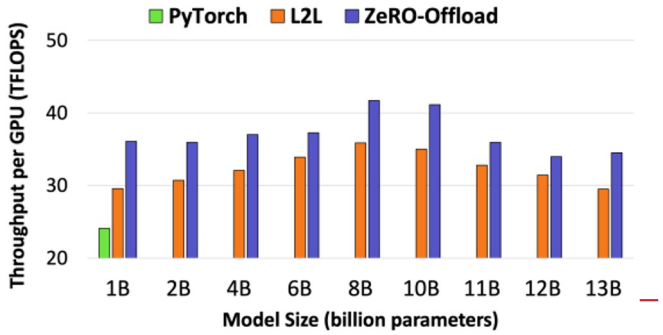


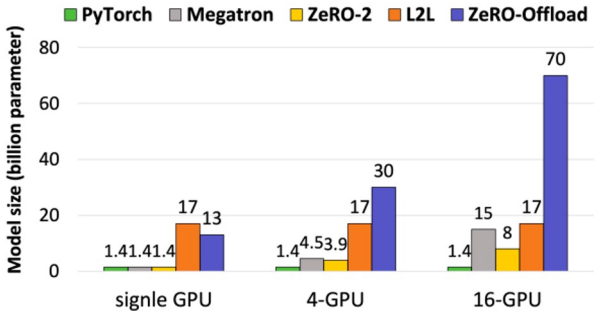
Figure 6: Delayed parameter update during the training process.

Memory offloading

ZeRO-offload: https://arxiv.org/abs/2101.06840

- Offload in parallel with computation
- Use gradient checkpointing
- Delayed parameter update



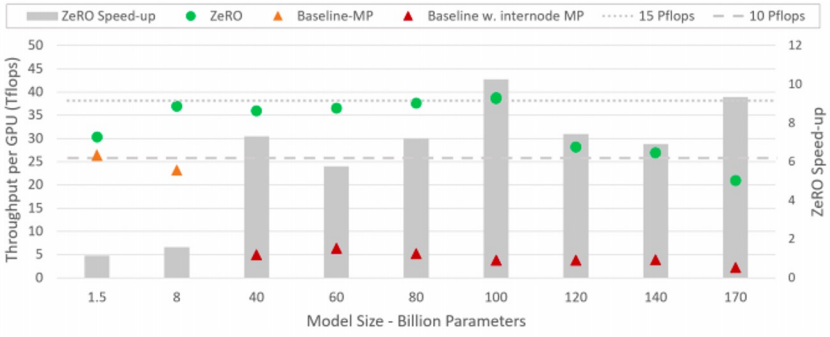


33

DeepSpeed / ZeRO

ZeRO: https://arxiv.org/pdf/1910.02054v3.pdf

- Combines sharded DPP and offload
- … and some tensor parallelism
- … and a ton of hacks



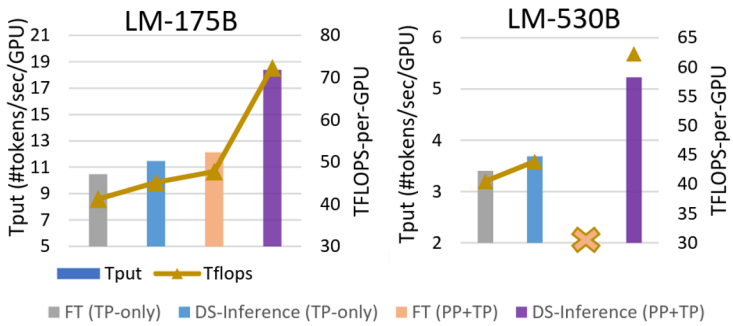
ĐẠI HỌC BÁCH K
HANOI UNIVERSITY OF SCIENC

34

DeepSpeed Inference

Paper: https://arxiv.org/abs/2207.00032

- Same techniques, but for inference
- Offloading, tensor- & pipeline-parallel
- … and a ton of hacks



ZeRO

- **Multi-GPU strategies:**
- * Pipeline model-parallel – allocate layers on different GPUs
- * Sharded data-parallel – split optimizer state and/or parameters

- **Single GPU strategies:**
- * Small model – gradient checkpointing & virtual batch
- * Large model – optimizer state sharding (keep parameters on GPU)

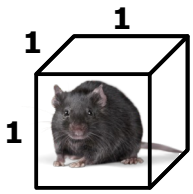
- **Implementations:**

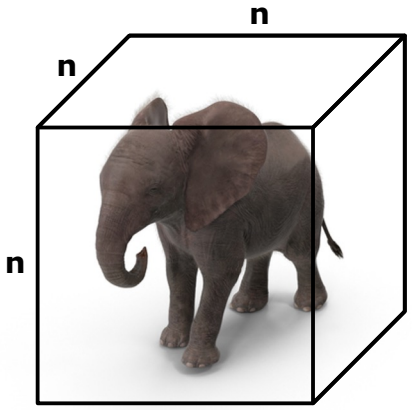
.DeepSpeed– sharded DP, offload, tensor parallelism, active development
.Offload – https://www.deepspeed.ai/news/2021/03/07/zero3-offload.html

.FSDP – most of DeepSpeed features with native PyTorch API
.Model-specific implementations– https://github.com/NVIDIA/Megatron-LM

The square-cube law of deep learning

Explainer: https://www.youtube.com/watch?v=f7KSfjv4Oq0





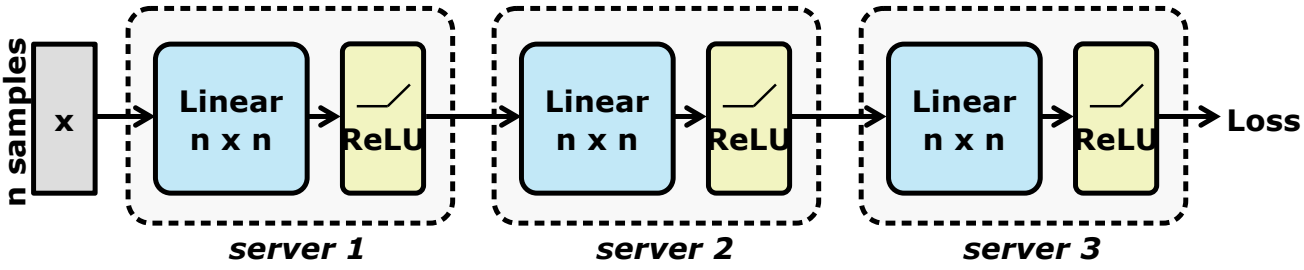
Surface area $O(n^2)$

Volume $O(n^3)$

37

The square-cube law of deep learning

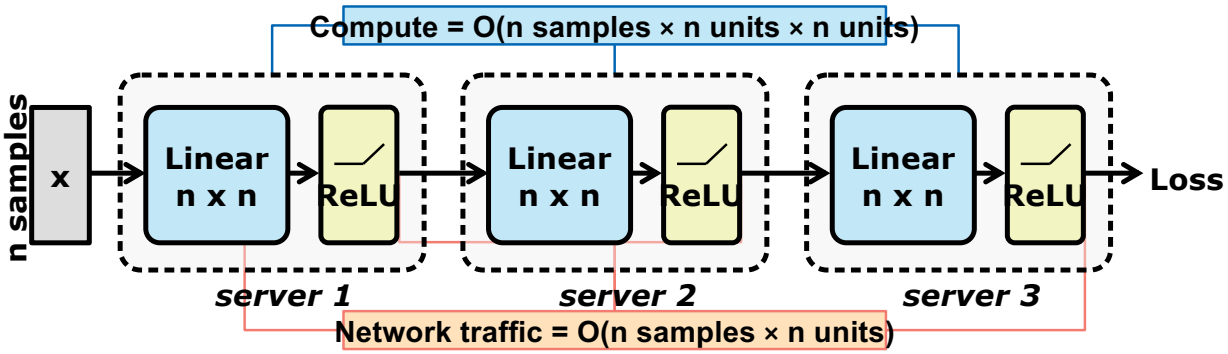
Explainer: https://www.youtube.com/watch?v=f7KSfjv4Oq0



38

The square-cube law of deep learning

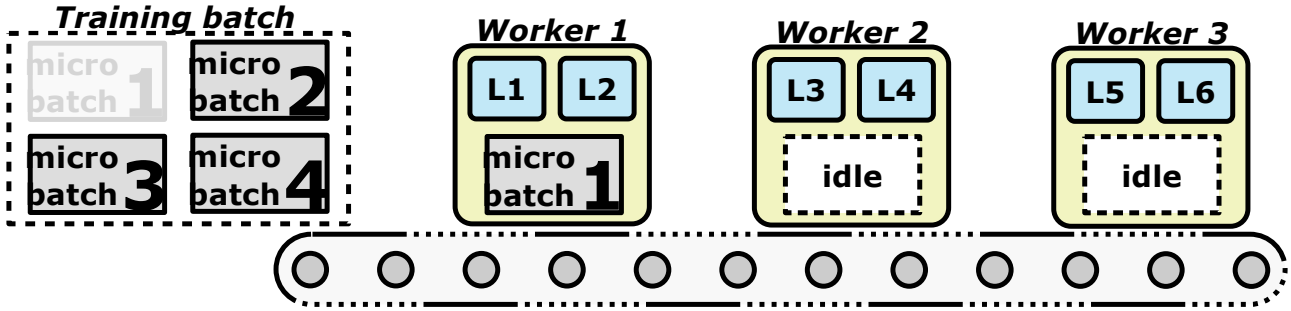
Explainer: https://www.youtube.com/watch?v=f7KSfjv4Oq0



39

The square-cube law of deep learning

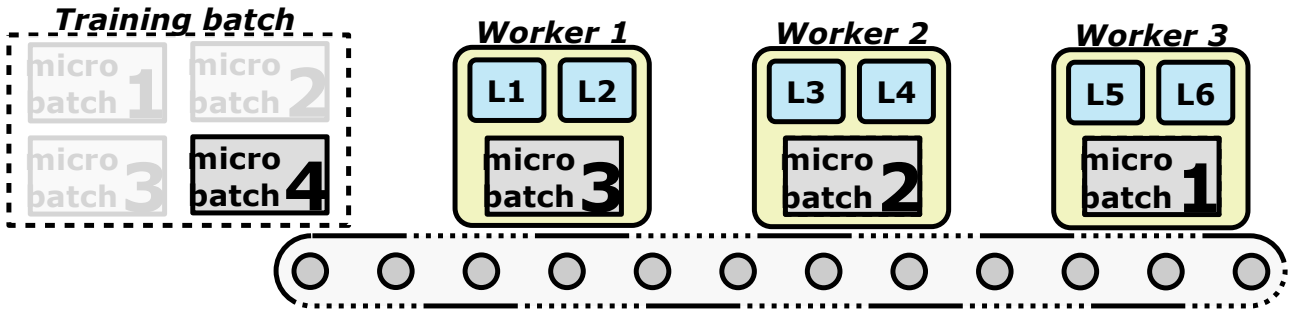
Explainer: https://www.youtube.com/watch?v=f7KSfjv4Oq0



40

The square-cube law of deep learning

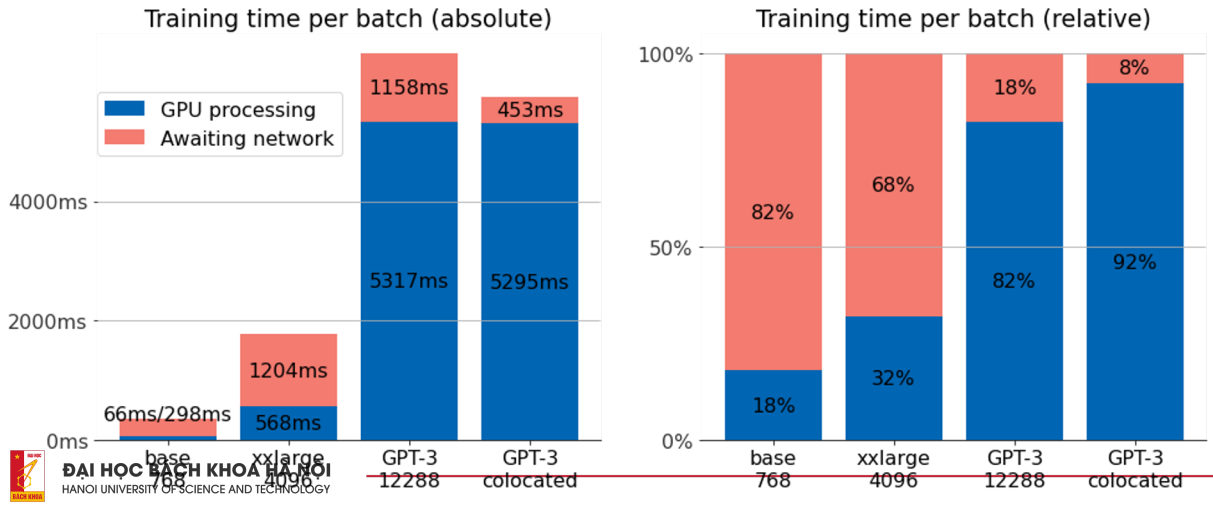
Explainer: https://www.youtube.com/watch?v=f7KSfjv4Oq0



41

The square-cube law of deep learning

Explainer: https://www.youtube.com/watch?v=f7KSfjv4Oq0



42

Petals

https://petals.ml

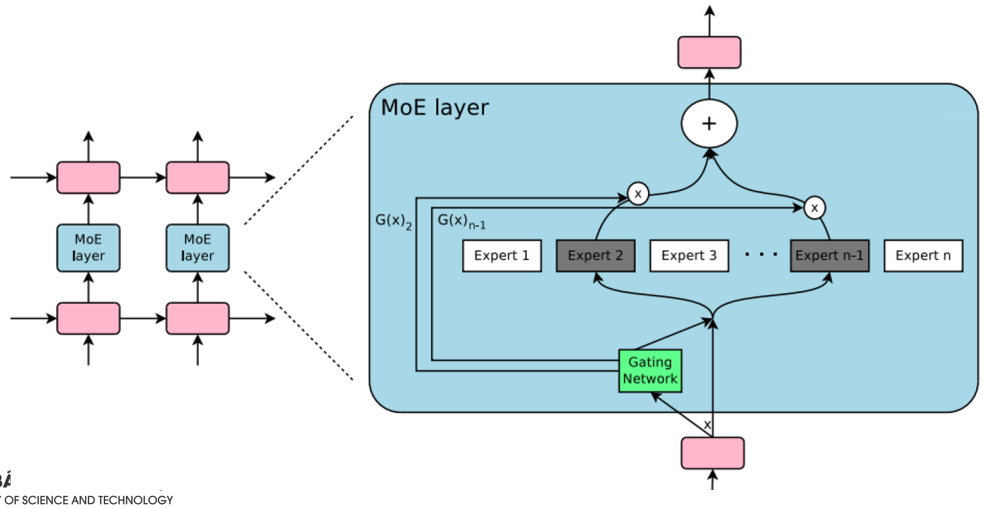
TL;DR: you can run 100B+ models over the internet, BitTorrent style

```
from petals import DistributedBloomForCausalLM

model = DistributedBloomForCausalLM.from_pretrained("bigscience/bloom-petals", tuning_mode="ptune",
# Embeddings & prompts are on your device, BLOOM blocks are distributed across the Internet

inputs = tokenizer("A cat sat", return_tensors="pt")["input_ids"]
outputs = model.generate(inputs, max_new_tokens=5)
print(tokenizer.decode(outputs[0]))  # A cat sat on a mat...

# Fine-tuning (updates only prompts or adapters hosted locally)
optimizer = torch.optim.AdamW(model.parameters())
for input_ids, labels in data_loader:
    outputs = model.forward(input_ids)
    loss = cross_entropy(outputs.logits, labels)
    optimizer.zero_grad()
    loss.backward()
    optimizer.step()
```

43

Expert Parallelism

Sparsely gated MoE: https://arxiv.org/pdf/1701.06538.pdf



ĐẠI HỌC BÁ
HANOI UNIVERSITY OF SCIENCE AND TECHNOLOGY

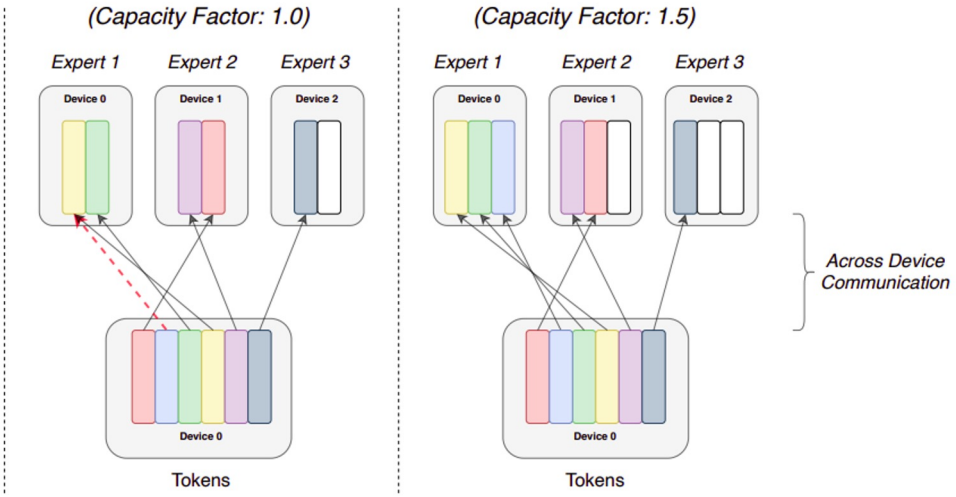
44

MoE Variant: Switch Transformer

Switch: https://arxiv.org/pdf/2101.03961.pdf

Terminology

- **Experts:** Split across devices, each having their own unique parameters. Perform standard feed-forward computation.
- **Expert Capacity:** Batch size of each expert. Calculated as
- (tokens_per_batch / num_experts) * capacity_factor
- **Capacity Factor:** Used when calculating expert capacity. Expert capacity allows more buffer to help mitigate token overflow during routing.

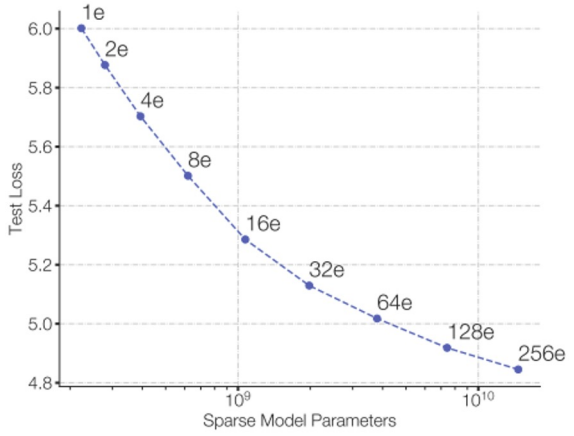


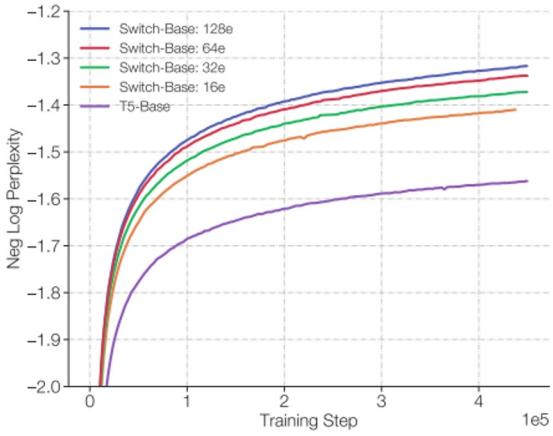
45

MoE Variant: Switch Transformer

Switch: https://arxiv.org/pdf/2101.03961.pdf

MLM pre-training objective [BERT-like]



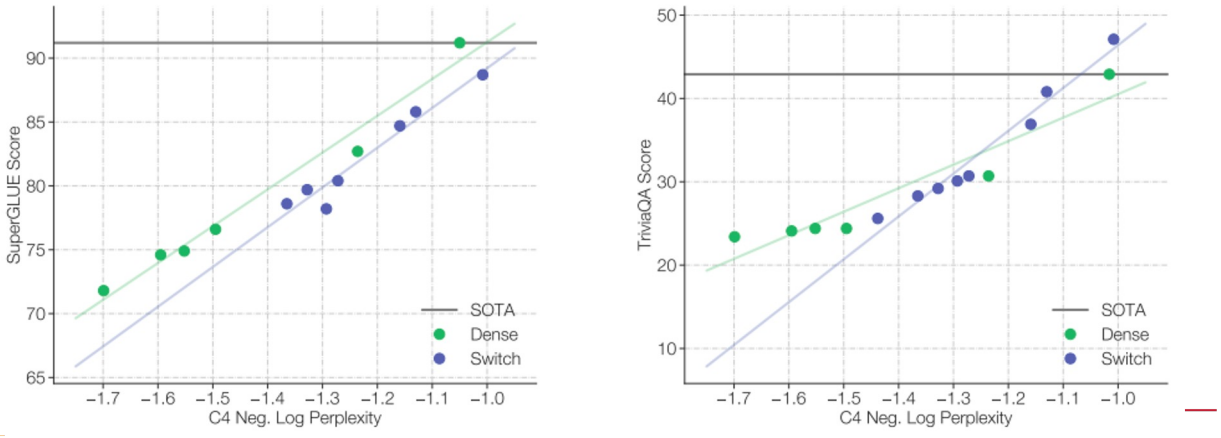


46

MoE Variant: Switch Transformer

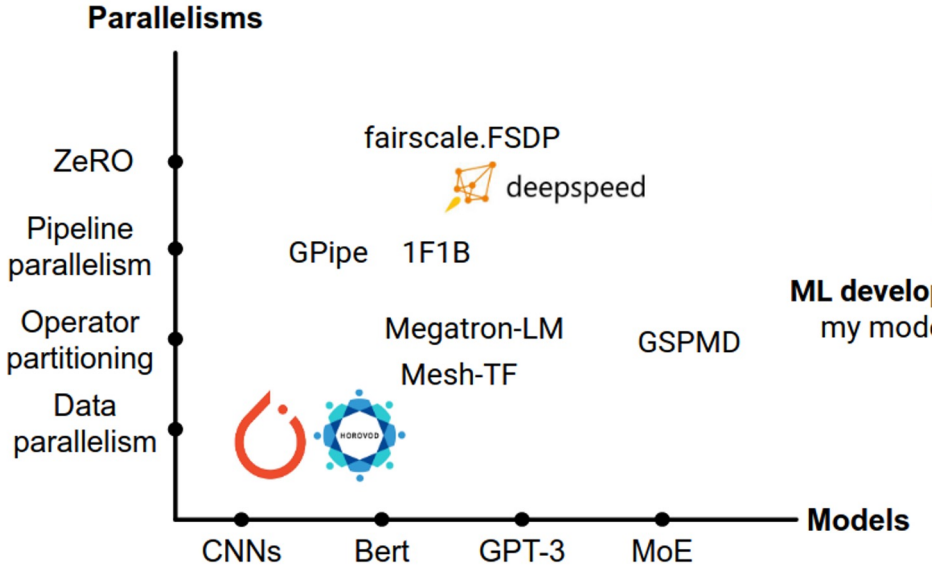
Switch: https://arxiv.org/pdf/2101.03961.pdf

Pre-training vs downstream quality



47

Automated parallelism



ML developer: which one is for my model and my cluster?

ĐẠI HỌC BÁCH KHOA HÀ NỘI
HANOI UNIVERSITY OF SCIENCE AND TECHNOLOGY

source: https://sites.google.com/view/icml-2022-big-model

48

Automated parallelism

Classic view

Data parallelism

Model parallelism

New view (this tutorial)

Inter-op parallelism

Intra-op parallelism

source: https://sites.google.com/view/icml-2022-big-model

49

Automated parallelism

Data parallelism

D partition → GPU GPU GPU GPU

$\theta, \nabla_L(\cdot), f(\cdot)$ replicate → GPU GPU GPU GPU

Model parallelism

$\theta, \nabla_L(\cdot)$ partition → GPU GPU GPU GPU

D ? → GPU GPU GPU GPU

$f(\cdot)$? → GPU GPU GPU GPU

$$\theta^{(t+1)} = f\left(\theta^{(t)}, \nabla_L\left(\theta^{(t)}, D^{(t)}\right)\right)$$

parameter — weight update (sgd, adam, etc.) — model (CNN, GPT, etc.) — data

50

Automated parallelism

Data and model parallelism

- Two pillars: **data** and **model**.
- ✅"Data parallelism" is general and precise.
- ❓"Model parallelism" is vague.
- ❓The view creates ambiguity for methods that neither partitions data nor the model computation.

New: Inter-op and Intra-op parallelism.

- Two pillars: **computational graph** and **device cluster**
- ✅This view is based on their computing characteristics.
- ✅This view facilitates the development of new parallelism methods.

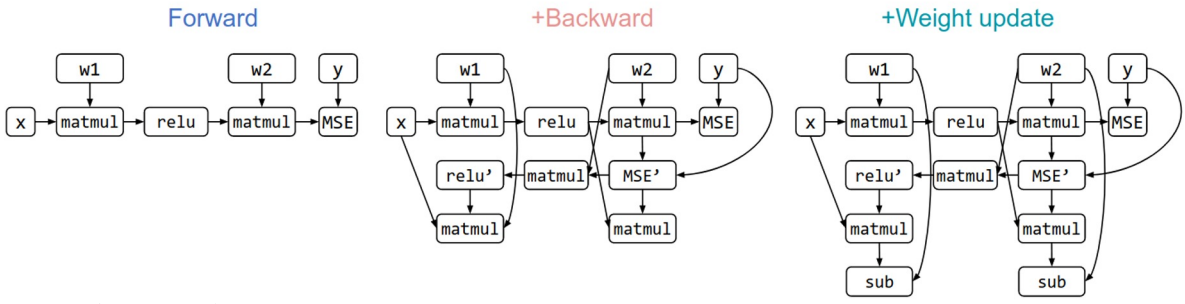
Automated parallelism

$$\theta^{(t+1)} = f\left(\theta^{(t)}, \nabla_L\left(\theta^{(t)}, D^{(t)}\right)\right)$$

$$L = \mathrm{MSE}(w_2 \cdot \mathrm{ReLU}(w_1 x),\ y) \quad \theta = \{w_1, w_2\},\ D = \{(x, y)\}$$

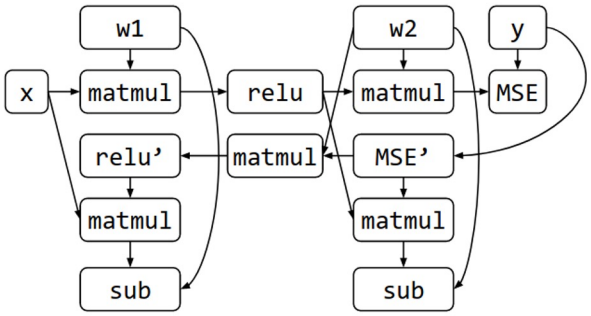
$$f(\theta, \nabla_L) = \theta - \nabla_L$$

Operator / its output tensor ⟶ Data flowing direction

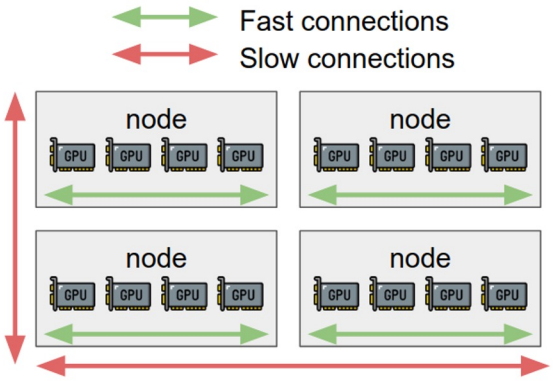


Automated parallelism

Compute graph



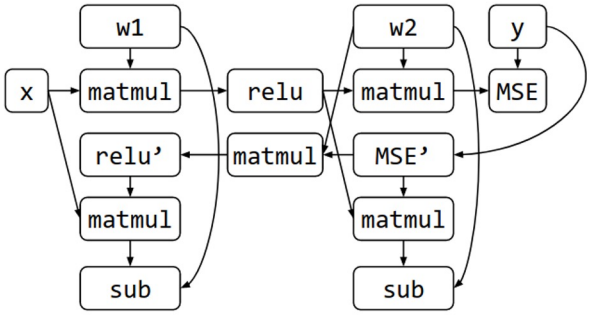
Device cluster

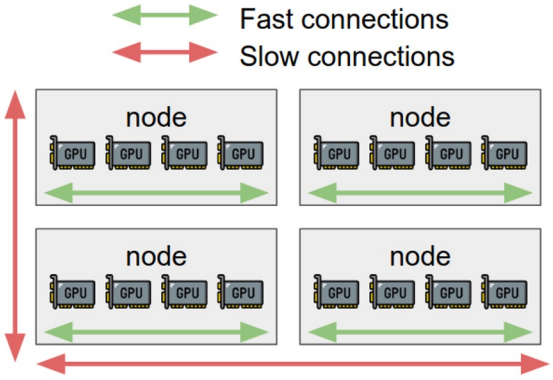


53

Automated parallelism

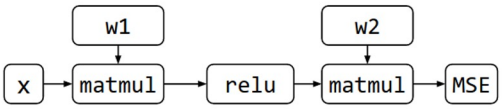
Q: How to partition the graph on the device cluster?



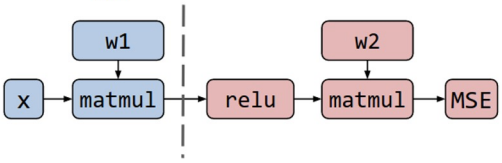


54

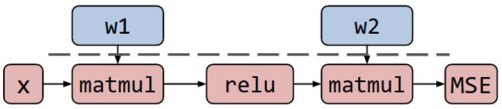
Automated parallelism



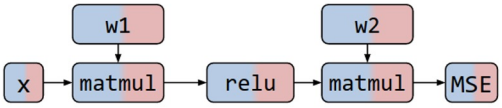
Strategy 1



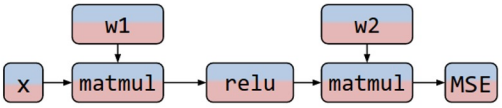
Strategy 2



Strategy 3

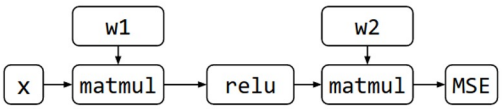


Strategy 4

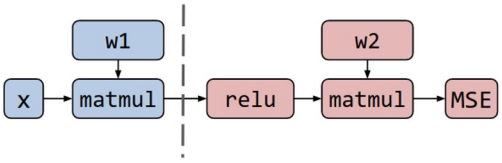


55

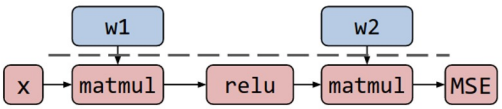
Automated parallelism



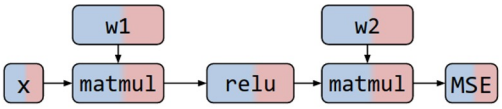
Strategy 1



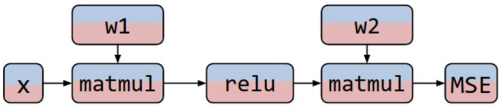
Strategy 2



Strategy 3

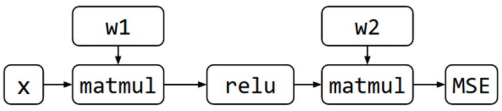


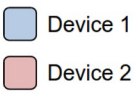
Strategy 4



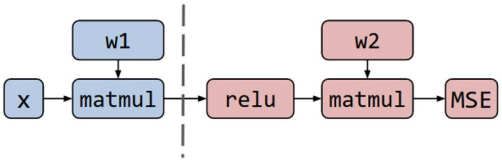
56

Automated parallelism

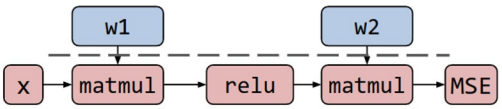




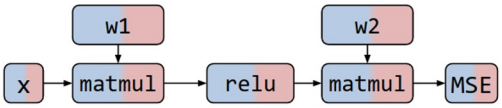
Pipeline MP



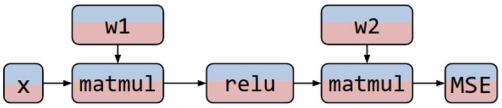
DP with offloading or PS



Tensor-parallel v1

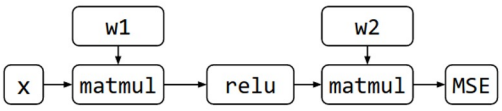


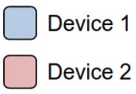
Tensor-parallel v2



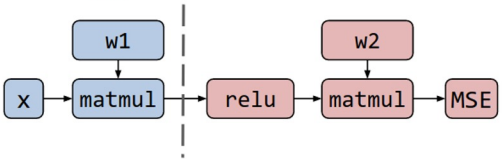
57

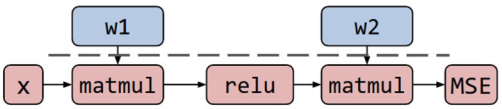
Automated parallelism



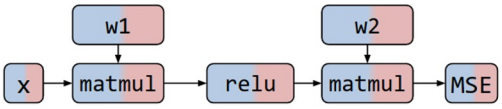


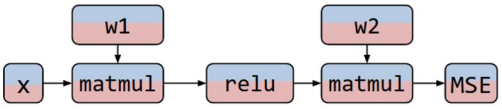
Pipeline MP **Inter-op parallelism** DP with offloading or PS





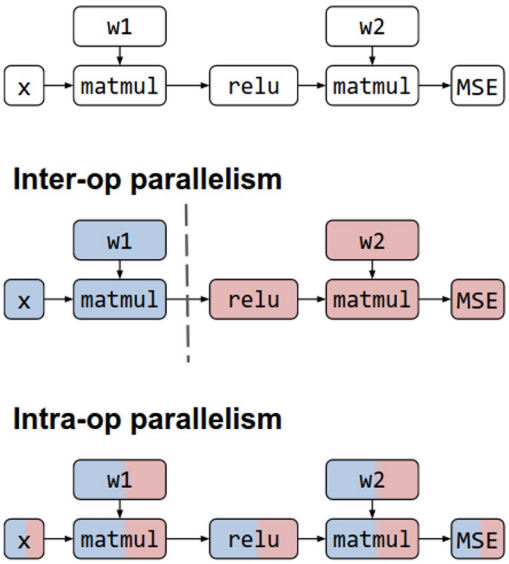
Tensor-parallel v1 **Intra-op parallelism** Tensor-parallel v2





58

Automated parallelism



Device 1

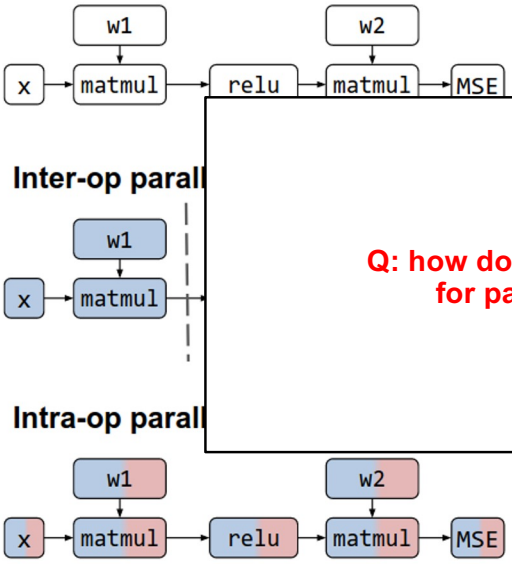
Device 2

Trade-off

	Inter-operator Parallelism	Intra-operator Parallelism
Communication	Less	More
Device Idle Time	More	Less

59

Automated parallelism



Device 1

Device 2

Q: how do we find the best strategy for partitioning the graph?

ntra-operator Parallelism

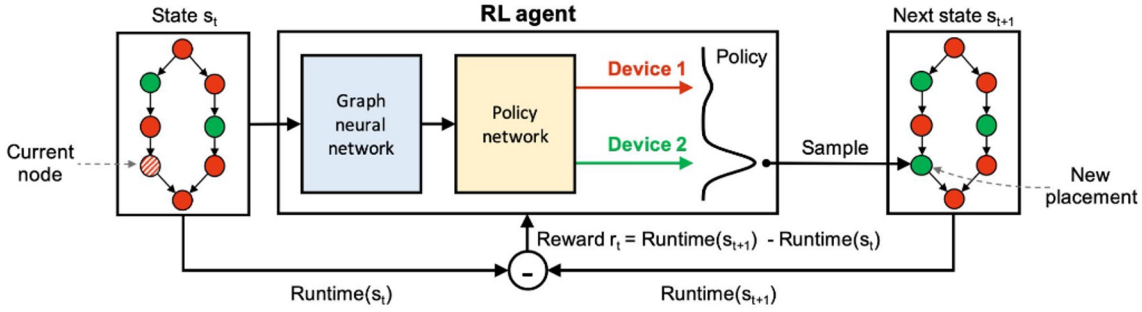
More

Less

60

RL-based partitioning

State: Device assignment plan for a computational graph.
Action: Modify the device assignment of a node.
Reward: Latency difference between the new and old placements.
Trained with **policy gradient** algorithm.



Addanki, Ravichandra, et al. "Placeto: Learning generalizable device placement algorithms for distributed machine learning." NeurIPS 2019.

61

Optimization-based partitioning

Integer Linear Programming:

Variable: Decision variable vector for each operator, representing device assignment.

Minimize: Maximum finishing time of all operators.

Constraint: Execution dependency & memory capacity of each device.

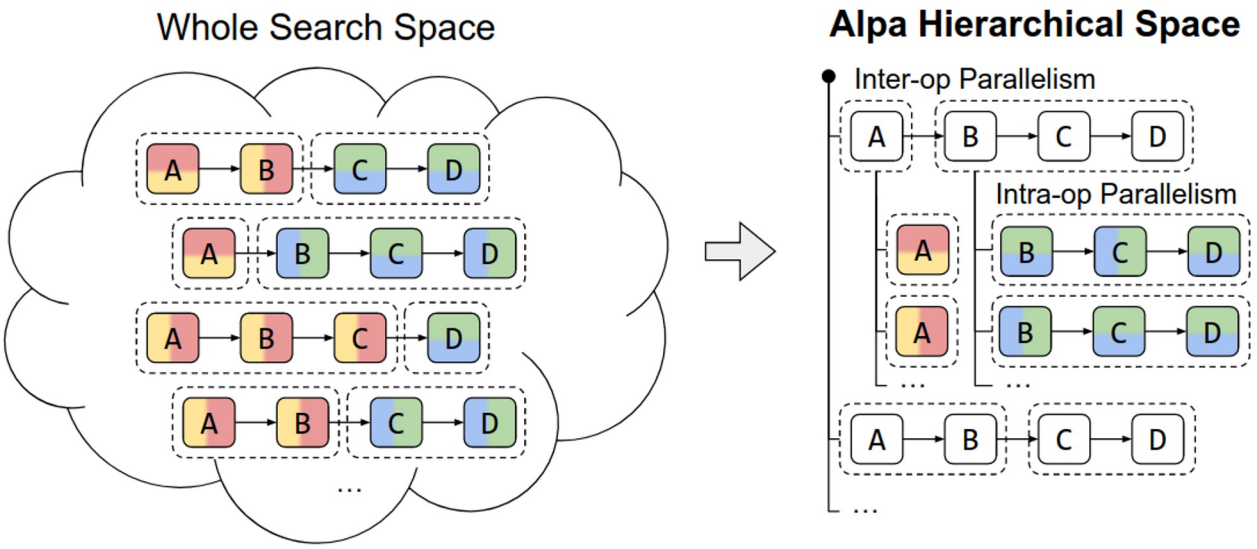
$$
\begin{aligned}
\min \quad & \text{TotalLatency} \\
\text{s.t.} \quad & \textstyle\sum_{i=0}^{k} x_{vi} = 1 \\
& \text{subgraph } \{v \in V : x_{vi} = 1\} \text{ is contiguous} \\
& M \geq \textstyle\sum_v m_v \cdot x_{vi} \\
& \text{CommIn}_{ui} \geq x_{vi} - x_{ui} \\
& \text{CommOut}_{ui} \geq x_{ui} - x_{vi} \\
& \text{TotalLatency} \geq \text{Latency}_v \\
& \text{SubgraphStart}_i \geq \text{Latency}_v \cdot \text{CommIn}_{vi} \\
& \text{SubgraphFinish}_i = \text{SubgraphStart}_i + \textstyle\sum_v \text{CommIn}_{vi} \cdot c_v \\
& \qquad + \textstyle\sum_v x_{vi} \cdot p_v^{\text{acc}} + \textstyle\sum_v \text{CommOut}_{vi} \cdot c_v \\
& \text{Latency}_v \geq x_{v0} \cdot p_v^{\text{cpu}} \\
& \text{Latency}_v \geq x_{v0} \cdot p_v^{\text{cpu}} + \text{Latency}_u \\
& \text{Latency}_v \geq x_{vi} \cdot \text{SubgraphFinish}_i \\
& x_{vi} \in \{0, 1\}
\end{aligned}
$$

Tarnawski, Jakub M., et al. "Efficient algorithms for device placement of dnn graph operators." NeurIPS 2020.

62

Alpa: optimization-based + reduced search space

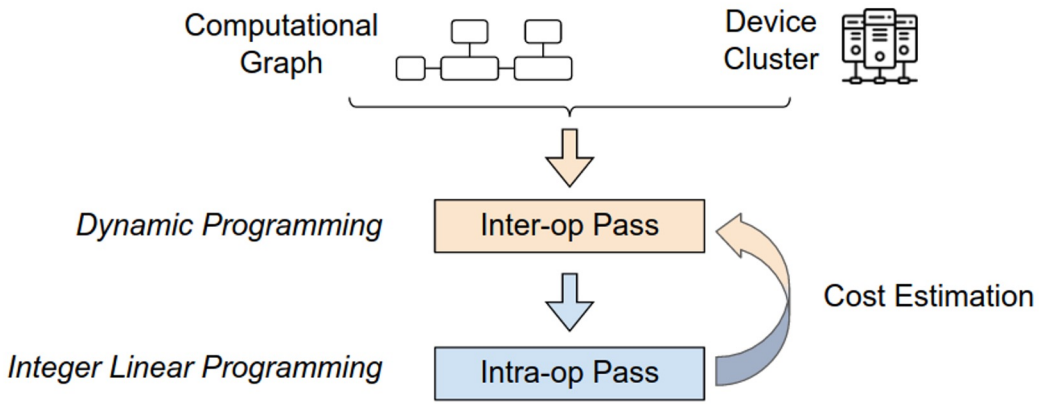
https://arxiv.org/abs/2201.12023



63

Alpa: optimization-based + reduced search space

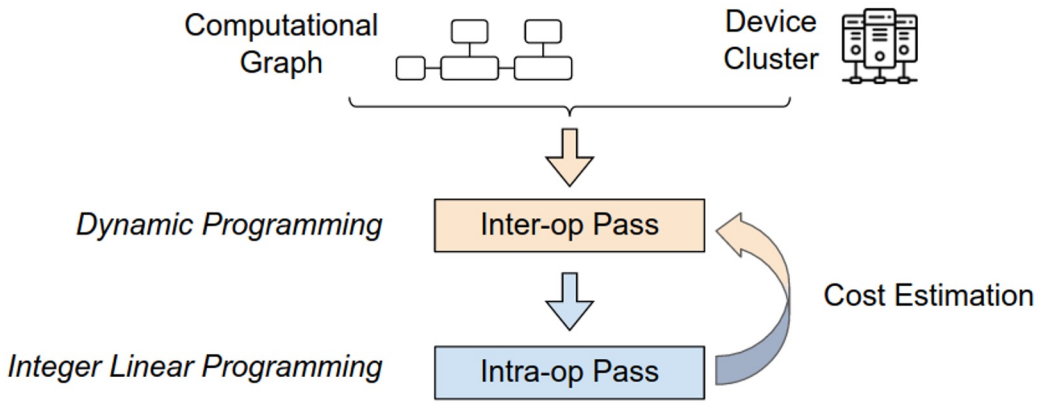
https://arxiv.org/abs/2201.12023



64

Alpa: optimization-based + reduced search space

https://arxiv.org/abs/2201.12023



More details of each pass:
https://sites.google.com/view/icml-2022-big-model

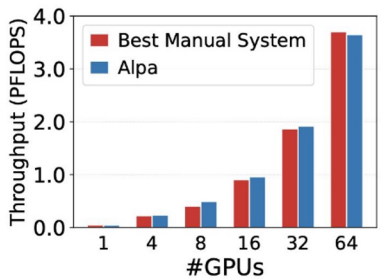
65

Alpa: optimization-based + reduced search space

https://arxiv.org/abs/2201.12023

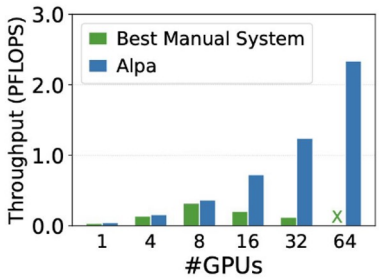
(benchmarks on AWS V100)

GPT (up to 39B)



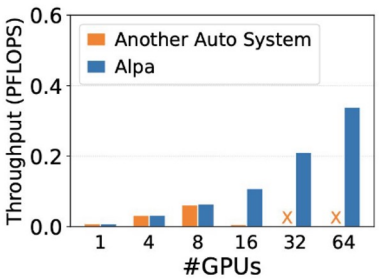
Match specialized manual systems.

GShard MoE (up to 70B)



Outperform the manual baseline by up to 8x.

Wide-ResNet (up to 13B)



Generalize to models without manual plans.

66

Alpa: optimization-based + reduced search space

- Not the first algorithm for auto-parallelism…
  but the first one that is usable* *(* - most of the time)*

```
# Define the training step. The body of this function is the same as the
# ``train_step`` above. The only difference is to decorate it with
# ``alpa.paralellize``.

@alpa.parallelize
def alpa_train_step(state, batch):
    def loss_func(params):
        out = state.apply_fn(params, batch["x"])
        loss = jnp.mean((out - batch["y"])**2)
        return loss

    grads = jax.grad(loss_func)(state.params)
    new_state = state.apply_gradients(grads=grads)
    return new_state

# Test correctness
actual_state = alpa_train_step(state, batch)
assert_allclose(expected_state.params, actual_state.params, atol=5e-3)
```

auto best strategy

works in jax

https://arxiv.org/abs/2201.12023

67

Example configuration

Several GPU w/ 24GB memory | 128GB system memory | 16GBps interconnect

16GB model and optimizer, 128GB activations (batch 32) → **grad accumulation**

16GB model and optimizer, 16GB activations (batch 1) → **grad checkpointing**

32GB model and optimizer, 1GB activations → **it depends…**

DDP + offloading	**FSDP (ZeRO)**	**Pipeline-parallel**	**Tensor-parallel**
e.g. if too few GPUs for other methods	*no custom model code, best for large batches*	*communication-efficient sequential model*	*minimal latency non-symmetric model*

Mix and match: TP within one server, minimal PP between servers, DDP between groups
Parallel code: manual (e.g. Megatron-LM) vs automated (alpa, FSDP, tensor_parallel)
Unconventional hardware: hivemind, petals, varuna, etc

68

Example configuration

Several GPU w/ 24GB memory | 128GB system memory | 16GBps interconnect

16GB model and optimizer, 128GB activations (batch 32) → **grad accumulation**

16GB model and optimizer, 16GB activations (batch 1) → **grad checkpointing**

32GB model and optimizer, 1GB activations → it depends…

If the model does not fit, you can also quantize it into submission!
(more on model compression in a future lecture)

Unconventional hardware: Nervana, petals, Tarana, etc